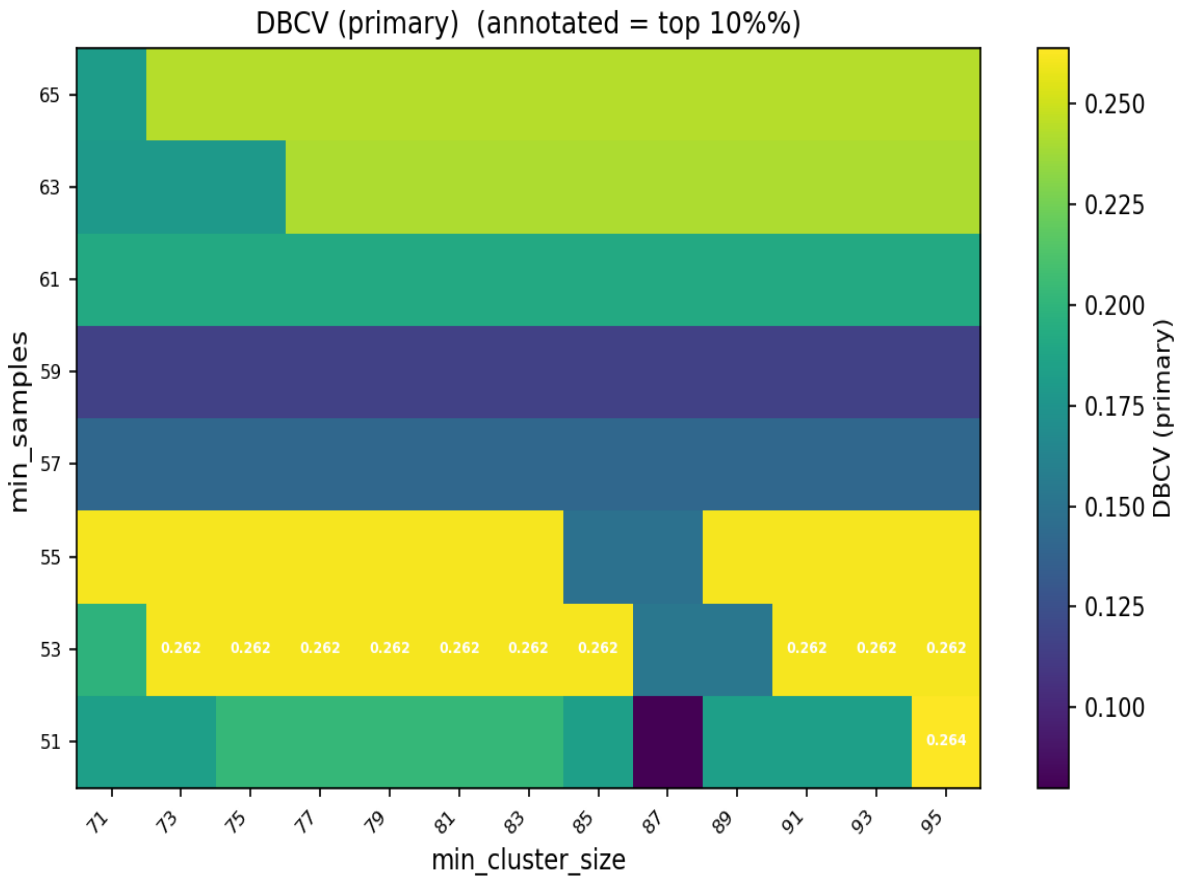
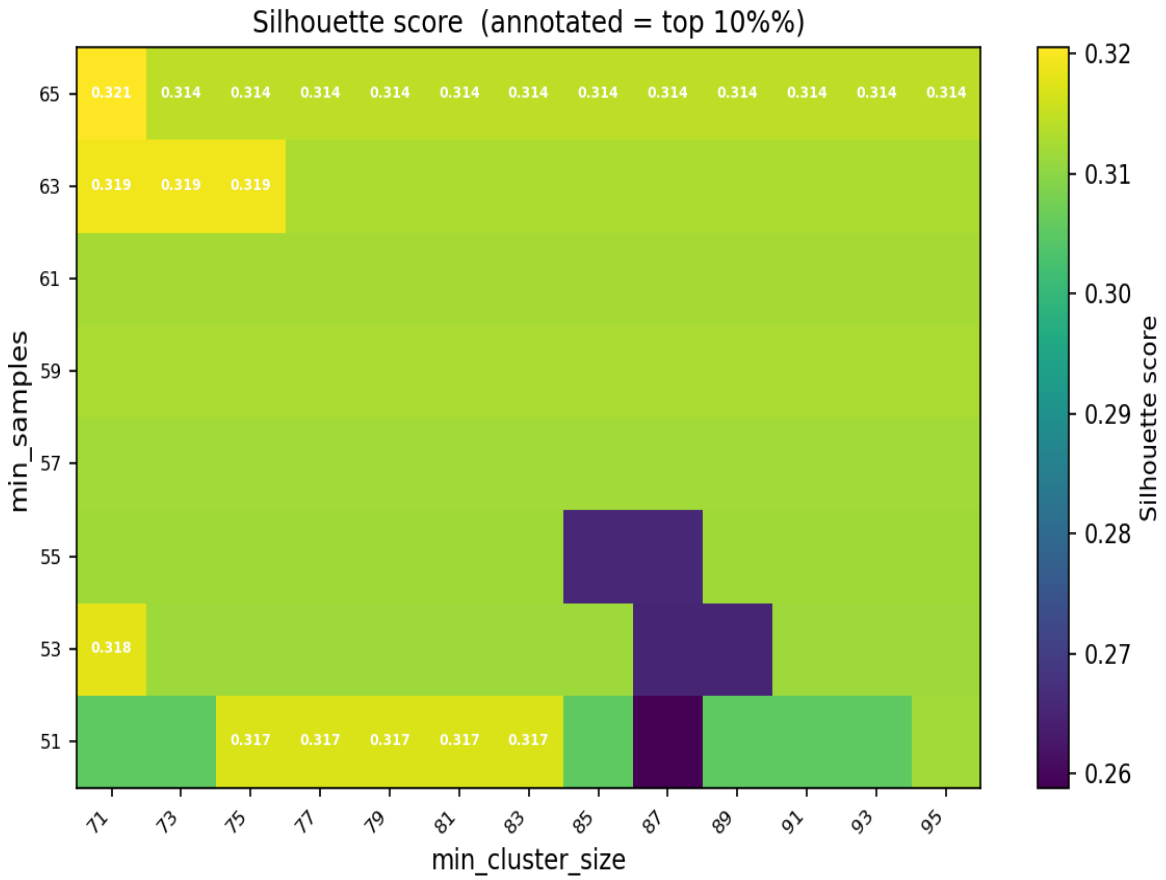
FragTree — Fine Grid Search Report

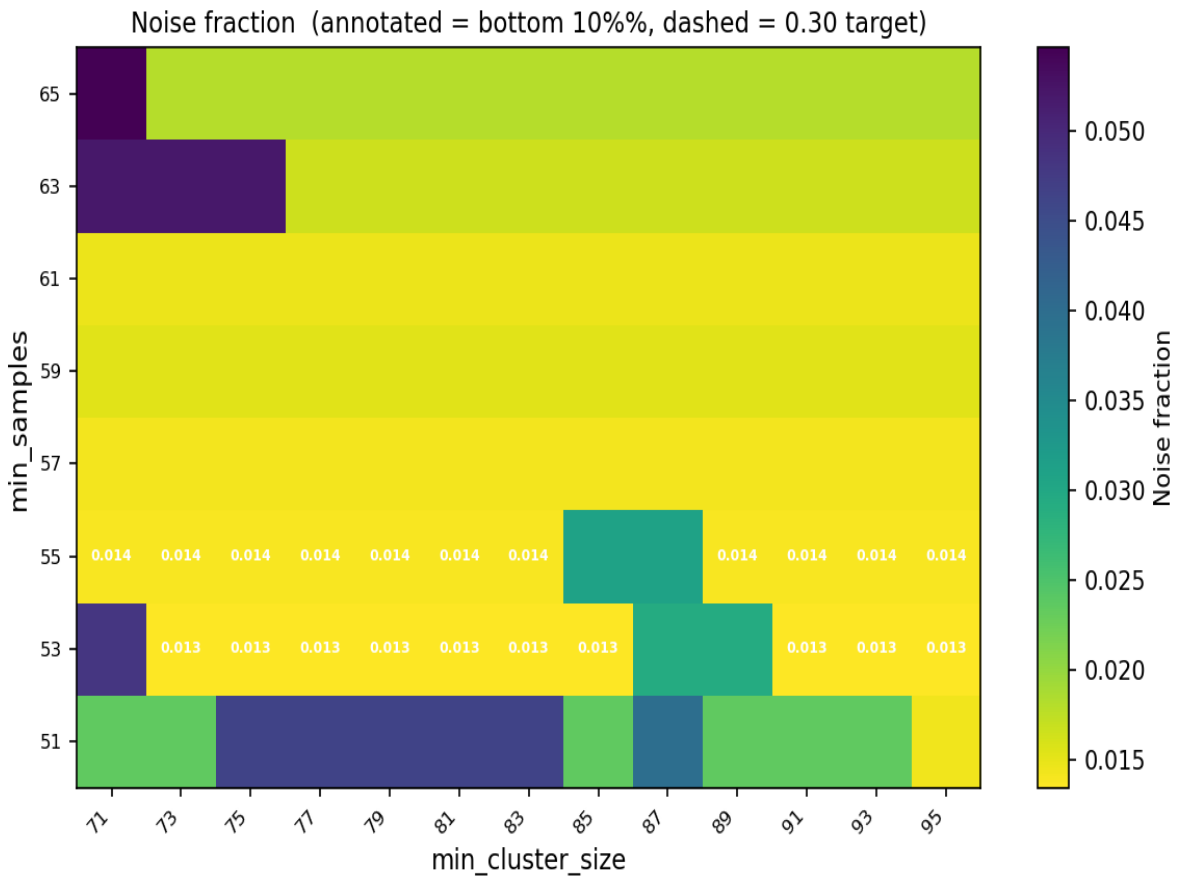
Generated: 2026-04-14 13:38:36 | n_molecules: 30,494

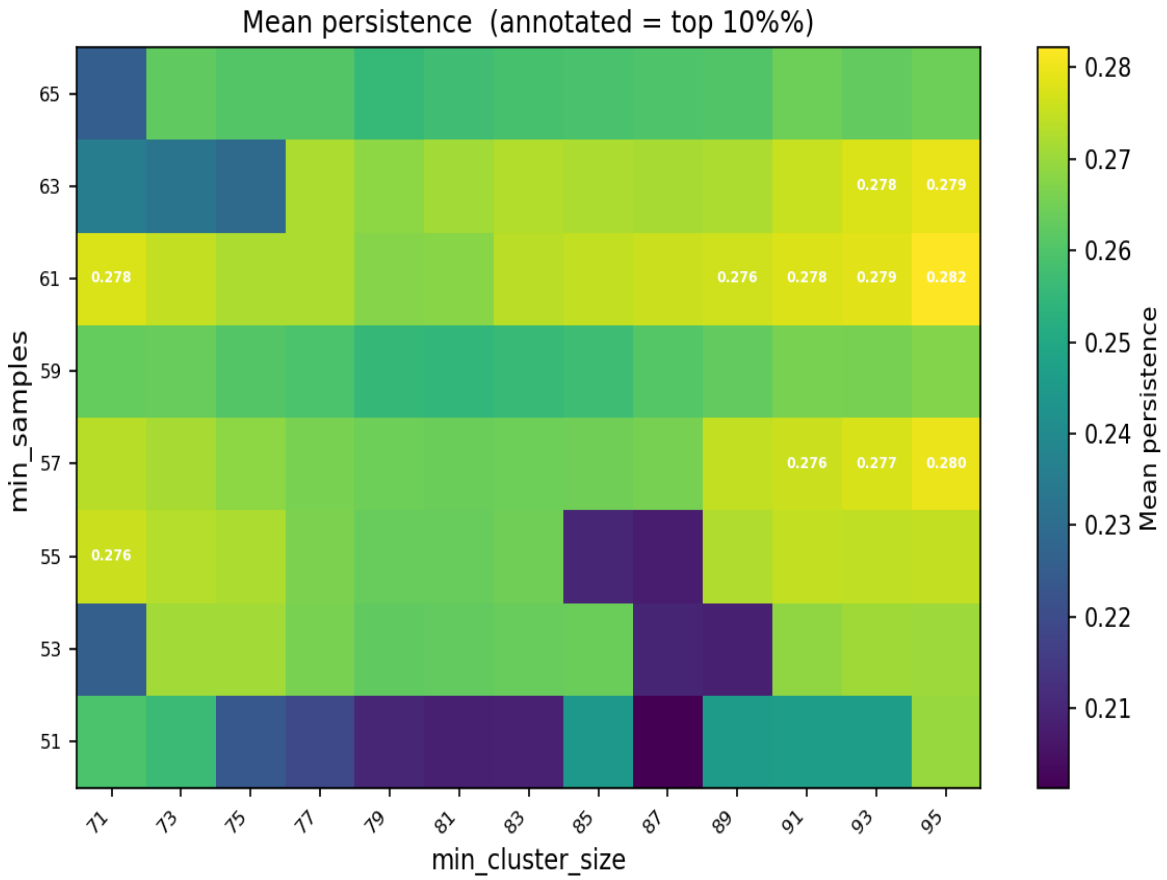
Parameter Sweep Heatmaps

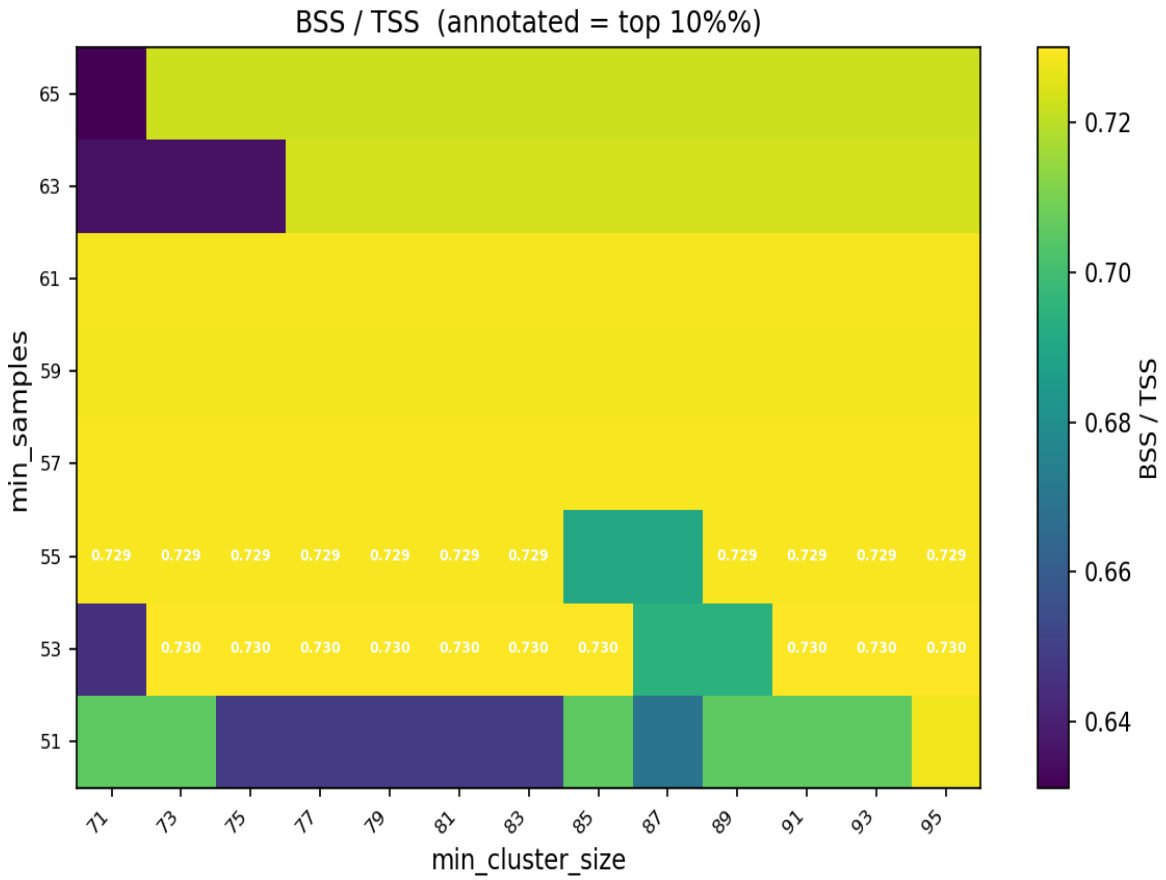
Each cell shows the metric value for a (min_cluster_size, min_samples) pair. Maximize DBCV, silhouette, mean persistence, and BSS/TSS. Minimize noise fraction (keep below 0.30). Cells with values in the top 10% are annotated with their score.




Silhouette score (annotated = top 10%%)
min_samples
min_cluster_size
Silhouette score
65
63
61
59
57
55
53
51
71
73
75
77
79
81
83
85
87
89
91
93
95
0.32
0.31
0.30
0.29
0.28
0.27
0.26
0.321
0.314
0.319
0.318
0.317


Noise fraction (annotated = bottom 10%%, dashed = 0.30 target)
min_samples
65
63
61
59
57
55
53
51
0.014
0.013
min_cluster_size
71
73
75
77
79
81
83
85
87
89
91
93
95
Noise fraction
0.050
0.045
0.040
0.035
0.030
0.025
0.020
0.015


Mean persistence (annotated = top 10%%)
min_samples
min_cluster_size
Mean persistence
65
63
61
59
57
55
53
51
71
73
75
77
79
81
83
85
87
89
91
93
95
0.278
0.279
0.278
0.276
0.278
0.279
0.282
0.276
0.277
0.280
0.276
0.28
0.27
0.26
0.25
0.24
0.23
0.22
0.21


BSS / TSS (annotated = top 10%%)
min_samples
65
63
61
59
57
55
53
51
0.729
0.730
71
73
75
77
79
81
83
85
87
89
91
93
95
min_cluster_size
0.72
0.70
0.68
0.66
0.64
BSS / TSS

Top-10% Summary Table

A parameter pair is included if it falls in the top 10% of ANY metric (or bottom 10% for noise fraction, where lower is better). Sorted by DBCV descending.

cs	ms	DBCV	Silhouette	Noise frac	Mean persistence	BSS/TSS	N clusters
95	51	0.2639	0.3120	0.014	0.2697	0.7279	7
73	53	0.2618	0.3115	0.013	0.2709	0.7300	7
75	53	0.2618	0.3115	0.013	0.2709	0.7300	7
77	53	0.2618	0.3115	0.013	0.2659	0.7300	7
79	53	0.2618	0.3115	0.013	0.2625	0.7300	7
81	53	0.2618	0.3115	0.013	0.2629	0.7300	7
83	53	0.2618	0.3115	0.013	0.2635	0.7300	7
85	53	0.2618	0.3115	0.013	0.2640	0.7300	7
91	53	0.2618	0.3115	0.013	0.2689	0.7300	7
93	53	0.2618	0.3115	0.013	0.2705	0.7300	7
95	53	0.2618	0.3115	0.013	0.2704	0.7300	7
71	55	0.2614	0.3117	0.014	0.2757	0.7291	7
73	55	0.2614	0.3117	0.014	0.2733	0.7291	7
75	55	0.2614	0.3117	0.014	0.2723	0.7291	7
77	55	0.2614	0.3117	0.014	0.2662	0.7291	7
79	55	0.2614	0.3117	0.014	0.2632	0.7291	7
81	55	0.2614	0.3117	0.014	0.2635	0.7291	7
83	55	0.2614	0.3117	0.014	0.2647	0.7291	7
89	55	0.2614	0.3117	0.014	0.2726	0.7291	7
91	55	0.2614	0.3117	0.014	0.2749	0.7291	7
93	55	0.2614	0.3117	0.014	0.2745	0.7291	7
95	55	0.2614	0.3117	0.014	0.2748	0.7291	7
73	65	0.2437	0.3143	0.018	0.2625	0.7225	7
75	65	0.2437	0.3143	0.018	0.2606	0.7225	7
77	65	0.2437	0.3143	0.018	0.2605	0.7225	7
79	65	0.2437	0.3143	0.018	0.2554	0.7225	7
81	65	0.2437	0.3143	0.018	0.2574	0.7225	7

83	65	0.2437	0.3143	0.018	0.2586	0.7225	7
85	65	0.2437	0.3143	0.018	0.2589	0.7225	7
87	65	0.2437	0.3143	0.018	0.2599	0.7225	7
89	65	0.2437	0.3143	0.018	0.2604	0.7225	7
91	65	0.2437	0.3143	0.018	0.2643	0.7225	7
93	65	0.2437	0.3143	0.018	0.2629	0.7225	7
95	65	0.2437	0.3143	0.018	0.2644	0.7225	7
93	63	0.2415	0.3130	0.017	0.2777	0.7233	7
95	63	0.2415	0.3130	0.017	0.2795	0.7233	7
75	51	0.2027	0.3171	0.046	0.2238	0.6487	11
77	51	0.2027	0.3171	0.046	0.2191	0.6487	11
79	51	0.2027	0.3171	0.046	0.2102	0.6487	11
81	51	0.2027	0.3171	0.046	0.2083	0.6487	11
83	51	0.2027	0.3171	0.046	0.2087	0.6487	11
71	53	0.1992	0.3181	0.048	0.2257	0.6454	11
71	61	0.1917	0.3121	0.015	0.2777	0.7284	7
89	61	0.1917	0.3121	0.015	0.2765	0.7284	7
91	61	0.1917	0.3121	0.015	0.2780	0.7284	7
93	61	0.1917	0.3121	0.015	0.2785	0.7284	7
95	61	0.1917	0.3121	0.015	0.2822	0.7284	7
71	65	0.1814	0.3205	0.055	0.2254	0.6313	11
71	63	0.1793	0.3191	0.052	0.2355	0.6356	11
73	63	0.1793	0.3191	0.052	0.2324	0.6356	11
75	63	0.1793	0.3191	0.052	0.2293	0.6356	11
91	57	0.1414	0.3119	0.014	0.2757	0.7289	7
93	57	0.1414	0.3119	0.014	0.2773	0.7289	7
95	57	0.1414	0.3119	0.014	0.2797	0.7289	7